

Harsh Mani Tripathi

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Education

Dr BR Ambedkar National Institute of Technology, Jalandhar

B.Tech in Electronics and Communication Engineering, CGPA: 8.66/10

Aug 2023 – 2027

Jalandhar, India

Achievements

Competitive Programming (LeetCode): Guardian rating, peak **2202** (top ~1% globally) – View Profile

Competitive Programming (Codeforces): Expert rating, peak **1631** – View Profile

Google's The Big Code 2026: Ranked **Top 1500** among a national pool of participants in Google's flagship coding competition.

Nebula Nexus Hackathon 2025: 1st Runner-Up, Best Tech – built and shipped a working prototype under a competitive 24-hour deadline.

Freelance Delivery: Engaged as a Freelance Developer for ServiceNow, delivering client-facing software work independently.

Experience

Accenture

Custom Software Engineering Associate

- Built a production **FastAPI** backend for an AI-powered Workday monitoring system – polling REST APIs on a schedule and using a fine-tuned **Ollama** LLM to auto-diagnose failures, surfaced through a Human-in-the-Loop dashboard for one-click fixes.
- Implemented APScheduler-driven job orchestration, OAuth 2.0 token auto-refresh, and SQLite-backed event persistence resilient to continuous scheduled load – hands-on experience wiring backend infrastructure directly to an LLM inference layer.
- Tech Stack:** Python (FastAPI), React.js (Vite), SQLite (SQLAlchemy), Ollama (Llama 3 / Mistral), APScheduler, REST APIs, OAuth 2.0

FocusMaster [GitHub]

AI & Backend Developer Intern

- Shipped a browser extension with a rules engine (scheduled, duration-based, manual modes) to block distracting sites, backed by a companion website (React.js, Node.js/Express, MongoDB) for schedule management and session analytics.
- Integrated **OpenAI's Whisper API** to parse voice commands into structured backend actions, reducing manual configuration during active sessions.
- Tech Stack:** Python, OpenAI Whisper API, React.js, Node.js/Express.js, MongoDB, REST APIs

Projects

EKA – Offline Enterprise RAG Assistant | GitHub

- Built a production-oriented Retrieval-Augmented Generation backend that answers employee questions from internal documents, served through a **FastAPI** REST API and running entirely offline with no external API keys.
- Implemented a hybrid retrieval pipeline combining FAISS dense vector search with BM25 sparse keyword search to improve answer relevance over single-method retrieval.

Invoice Ingestion Agent | GitHub

- Automated end-to-end invoice processing with a locally hosted pipeline (n8n workflow orchestration, Ollama/Llama 3.2:3b for extraction, NocoDB for structured storage), removing manual data entry – directly applicable to Copilot-style operational-workflow automation.

Type-Short – Multilingual Text Summarizer | GitHub

- Built a **Flask** web app serving a **T5 transformer** for text summarization, integrated with the Google Translate API to deliver summaries in a user-selected language.
- Hands-on experience serving a transformer model behind a Python backend with multilingual output – directly relevant to scaling Gen AI features like TTS across languages.

FileFare – Full-Stack Digital Marketplace | GitHub

- Architected a full-stack marketplace (React 19/Vite, **Node.js/Express**, **MongoDB**) across a 5-model data layer (Users, Files, Purchases, Reviews, Withdrawals), with role-based dashboards, catalog search/filtering, and ratings.
- Implemented Stripe Payment Intents, JWT + bcrypt auth, and secure download tokens; hardened the REST API with Helmet, express-rate-limit, and CORS.

Technical Skills

Languages: Python, Java, C/C++, JavaScript, SQL

Backend & APIs: FastAPI, Flask, Django, Node.js/Express, RESTful API design, OAuth 2.0, microservices fundamentals

Databases: PostgreSQL, MongoDB, SQLite, MySQL, NocoDB

AI/ML & LLMs: Ollama (Llama 3 / Mistral), PyTorch, TensorFlow, Hugging Face Transformers, T5, FAISS, Scikit-learn, workflow automation (n8n)

Systems & Tools: UDP/TCP sockets, protocol design, AES-GCM/HMAC, multithreading, Docker, Git, GitHub, Azure, Postman, Jupyter